



Review Article

Artificial intelligence in veterinary and animal science: applications, challenges, and future prospects

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ARTICLE INFO

Keywords:

Artificial intelligence
Deep learning
Machine learning
Precision livestock farming
Veterinary diagnostics

ABSTRACT

Artificial intelligence (AI) is increasingly transforming animal and veterinary science through improved decision-making, predictive modeling, and automation. This review comprehensively covers applications of key AI technologies, including machine learning, deep learning, computer vision, natural language processing, robotics, and edge AI, in areas such as disease diagnosis, behavioral monitoring, multi-omics data integration, and precision livestock farming. It also highlights current limitations, including data fragmentation, high implementation costs, and ethical concerns. By addressing these challenges through interdisciplinary collaboration, standardized data systems, and the development of explainable and scalable AI tools, the field can advance toward more sustainable, efficient, and welfare-oriented practices. This review underscores the transformative potential of AI in achieving the goals of One Health and global food security while emphasizing the need for continued research, policy support, and equitable access to AI technologies.

1. Introduction

Artificial intelligence or AI refers to a comprehensive range of technologies that enable machines to imitate human intelligence. This field includes various sub-disciplines such as machine learning (ML), advanced machine learning known as deep learning (DL), natural language processing (NLP), and computer vision (CV) (Goodfellow et al., 2016). These capabilities are revolutionizing various sectors by introducing systems that absorb data, identify trends, and provide decision support with an unprecedented level of accuracy. In the realm of AI, ML and DL have attracted considerable interest due to their strong capacity to manage intricate data and their extensive use in various sectors, such as image and video recognition, forecasting, and automation (LeCun et al., 2015). This development is of great importance for animal and veterinary science, as the data in these fields are often large, disjointed, and difficult to interpret using traditional methods.

As AI techniques have evolved, their integration into the fields of life and agricultural science has brought about significant changes. In the field of animal science and veterinary medicine, where complex, often personalized data is critical, the ability of AI to process and interpret large amounts of data will transform the way professionals and researchers work with animals, and significantly affect animal health, welfare, and productivity (Liakos et al., 2018). AI has shown promise in simplifying complex processes, including the detection of health

problems, the prediction of animal productivity, and the monitoring of behavior, and could potentially provide solutions to the ongoing problems in these sectors.

In the field of animal science, AI is widely used in various fields, which are essential to increase productivity, improve animal welfare, and promote sustainability. A significant application of AI can be seen in the field of animal breeding and genetics, where ML algorithms help predict traits, assess genetic diversity, and improve breeding strategies by interpreting data (Morota et al., 2018). Through better genetic selection, AI helps improve beneficial traits, such as increased growth rates, better feed conversion, and improved resistance to disease. This in turn helps producers meet sustainability objectives and promote animal health (Chafai et al., 2023).

AI plays an essential role in the field of advanced livestock farming, known as precision livestock farming (PLF). PLF uses AI-powered technologies to improve feed management, animal health monitoring, and environmental regulation, minimizing waste and increasing efficiency (Tedeschi et al., 2021). For example, advanced predictive models analyze sensor data to monitor environmental factors and individual animal responses, enabling adjustments to feed, temperature, or hydration that simultaneously promote animal well-being and increase production (Jiang et al., 2023). Using CV and sensor technology, AI helps monitor, in addition to its effectiveness, the behavior of animals. This allows for the real-time monitoring of behaviors such as aggression,

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Received 18 January 2025; Received in revised form 4 April 2025; Accepted 7 April 2025

Available online 16 April 2025

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